Extremum Seeking Control of a Unicycle-like Robot Using the Transverse Function Approach

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***Abstract*—This paper addresses the problem of extremum seeking control for a planar mobile robot of unicycle type subject to nonholonomic constraints. The proposed approach is based on the transverse function framework, which enables the transformation of the nonholonomic system into a form suitable for control design. To ensure persistent excitation, an excitation component is injected directly into the transverse function, allowing for effective exploration of the objective function. The gradient information required for extremum seeking is obtained through a local parametric approximation computed over a sliding time window. The resulting control scheme enables the robot to converge toward the extremum of an unknown spatial field while respecting the system's kinematic constraints. Simulation results are provided to demonstrate the effectiveness of the proposed method.**

I. Introduction

Extremum seeking control (ESC) is an important area of control theory with significant practical relevance [1]. Its objective is to steer a dynamical system toward the minimum or maximum of an unknown performance function using only measurements of its value. This makes ESC particularly suitable for real-time optimization problems where accurate models are unavailable or the objective function can only be measured [2], [3].

In mobile robotics, ESC naturally arises in source-seeking tasks, where a robot must locate the extremum of a spatial scalar field representing quantities such as chemical concentration, temperature, or signal strength [4]. Since constructing a full environmental map is often impractical, control strategies based solely on local sensor feedback are especially valuable.

The problem becomes more challenging for robots subject to nonholonomic constraints [5], where classical gradient-following strategies cannot be directly applied. Various methods have been proposed to address this issue, including approaches based on oscillatory perturbations that extract directional information from measurements [6], as well as techniques rooted in differential geometry and Lie bracket motion generation [7], [8]. Extensions to time-varying fields have also been studied [9].

In this paper, we address the ESC problem by exploiting the concept of transverse functions introduced by Morin and Samson [10]–[12]. By augmenting the system with an auxiliary dynamic structure, the transverse function approach allows the original nonholonomic model to be reformulated into a representation that is more suitable for control design. In particular, this framework enables the synthesis of smooth control laws that ensure practical stabilization and tracking while respecting the inherent nonholonomic constraints of the system.

We consider the ESC problem for a unicycle-like mobile robot subject to nonholonomic constraints. The main contribution of this work is the use of a time-varying transverse function that simultaneously fulfills two roles. First, it serves as the principal component of the motion controller, enabling the system to overcome limitations imposed by the kinematic constraints. Second, it generates a perturbation signal that facilitates the recovery of local gradient information of the unknown objective function. Moreover, the proposed control structure allows independent regulation of the robot orientation, thereby extending the range of potential applications.

In contrast to the classical ESC framework, which relies on low-pass and high-pass filters to extract gradient information, we employ a local polynomial approximation of the objective function and formulate a regression problem over a sliding time window to estimate the gradient. Finally, simulation results are presented to demonstrate the performance of the proposed approach for different parameterizations of the transverse function.

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II. Control systems on SE(2)

A. Mathematical preliminaries

Consider the Lie group $G \simeq SE(2)$, whose elements are represented in vector form as $\forall g \in G,\ g = \begin{bmatrix} g_x & g_y & g_\theta \end{bmatrix}^\top \in \mathbb{R}^2 \times S^1$. The group operation corresponds to the composition of planar rigid-body transformations and, for all $g, h \in G$, is defined as

$$gh = g + T(g_\theta)h, \tag{1}$$

where

$$T(g_\theta) = \begin{bmatrix} R(g_\theta) & 0 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 3}, \tag{2}$$

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and $R(g_\theta) \in SO(2)$ denotes the planar rotation matrix. The neutral element is given by $e = 0 \in G$, and the inverse of an element $g \in G$ satisfies

$$g^{-1} = -T^\top(g_\theta)g.$$

For later use, define the following operators: the left translation $l_g : G \to G$, $l_g(h) = gh$, the right translation $r_g : G \to G$, $r_g(h) = hg$, and the conjugation map $\phi_g : G \to G$, $\phi_g(h) = ghg^{-1}$. In addition, to represent only two first components of the element on G, which live in $\mathbb{R}^2$, we introduce the following notation: $\forall g \in G$, $g_{xy} = [g_x \ g_y]^\top \in \mathbb{R}^2$.

Let the i-th column of the matrix $T(g_\theta)$ define a vector field X_i on G. These vector fields are left-invariant, i.e.,

$$X_i(gh) = dl_g(h)X_i(h), \tag{3}$$

where $dl_g(h) = \frac{d}{dh}l_g(h)$ denotes the derivative of the left translation. Assume that the vector fields X_i form a basis of the Lie algebra $\mathfrak{g}$ associated with the group G. In matrix form, this basis can be written compactly as $X(g) = T(g_\theta)$.

B. Kinematics on Lie group $G \simeq SE(2)$

Now, using the concept and notation presented above we can define two kinematics on Lie group G. The first kinematics correspond to omnidirectional unconstrained planar system which is given as:

$$\dot{z} = X(z)\nu, \tag{4}$$

where $z \in G$ stands for the system configuration and $\nu = [v_x \ v_y \ \omega]^\top \in \mathbb{R}^3$ is the input representing velocities in the body frame. The second kinematics concern unicycle-like planar vehicle and is defined as

$$\dot{g} = X(g)Cu, \tag{5}$$

where $C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}^\top$ and $u = [v_x \ \omega]^\top \in \mathbb{R}^2$.

III. Control design

A. Control problem

The essential control problem can be perceived as an ESC problem. The following is assumed with respect to the objective function μ:

Assumption 1. *The function $\mu : \mathbb{R}^2 \to \mathbb{R}$ is twice continuously differentiable and admits an isolated local maximum at $p^* \in \mathbb{R}^2$, i.e.,*

$$\nabla\mu(p^*) = 0, \quad \nabla^2\mu(p^*) \prec 0. \tag{6}$$

Assumption 2. *The initial condition $p(0)$ belongs to a neighborhood of p^*, i.e., $p(0) \in \mathcal{D}(p^*) \subset \mathbb{R}^2$, where $\mathcal{D}(p^*)$ denotes the domain of attraction of the maximum.*

Let $p \in \mathbb{R}^2$ denote the position of the unicycle (5). That is, p is contained in $g \in G$ and given by $p = g_{xy}$. The control objective is to design a feedback controller $u(g, \cdot)$ such that the closed-loop trajectory $p(t)$ converges to a vicinity of p^*, where the corresponding objective function value approaches μ^*, i.e.,

$$\lim_{t\to\infty} p(t) \in B_{\epsilon_p}(p^*), \quad \mu(p(t)) \in [\mu^* - \epsilon_\mu, \mu^*], \tag{7}$$

where $B_{\epsilon_p}(p^*)$ denotes a ball of radius ϵ_p, and $\epsilon_\mu > 0$ is a constant.

To solve this control problem, the nonholonomic constraints are addressed first. Subsequently, a method for local gradient estimation based on measurements is considered, since it is not available a priori.

B. Companion systems

In order to cope with the motion constraints resulting from the nonholonomy of the unicycle, we take advantage of planar holonomic kinematics and consider the following:

Assumption 3. *The systems* (4) *and* (5) *are coupled such that their states satisfy the constraint*

$$g = zf, \tag{8}$$

where $f \in G$ is a bounded function, satisfying so-called transversality conditions [10].

Function f briefly referred to as transverse function can be determined on S^1 taking into account control vector fields of kinematics (5) represented by $X(g)C$. According to the definition of Morin and Samson [11], transverse function $f : S^1 \to G$ is chosen in such a way that $\forall\alpha \in S^1 \operatorname{rank}\left[X(f(\alpha))C \ \frac{\partial f(\alpha)}{\partial\alpha}\right] = \dim G$ and $\forall\alpha \in S^1, \ f(\alpha) \in B_\epsilon$, where B_ϵ denotes a neighborhood of the neutral element with radius ϵ.

Moreover, to facilitate the gradient estimation discussed later, we assume that f explicitly depends on time and that its partial derivative $\frac{\partial f}{\partial t}$ is bounded. Consequently, its total time derivative satisfies $\dot{f} = \frac{\partial f}{\partial\alpha}\dot{\alpha} + \frac{\partial f}{\partial t}$. To facilitate the control design, we express the partial derivatives of f in the basis of the Lie algebra $\mathfrak{g}$ and introduce:

$$A_\alpha = X^{-1}(f)\frac{\partial f}{\partial\alpha}, \ A_t = X^{-1}(f)\frac{\partial f}{\partial t}. \tag{9}$$

As a result, we can write

$$\dot{f} = X(f)\left(A_\alpha\dot{\alpha} + A_t\right). \tag{10}$$

Calculating time derivative of both sides of (8), and using (4), (5) and (10) (details are omitted, as they can be found in [13], for example), one obtains the following map between u, $\dot{\alpha}$ and ν:

$$\nu = \mathrm{Ad}^X(f)\left(Cu - A_\alpha\dot{\alpha} - A_t\right), \tag{11}$$

where $\mathrm{Ad}^X(g) = X(e)^{-1}\mathrm{Ad}(g)X(e)$ is the adjoint representation expressed in the basis $X(e)$, where the adjoint operator $\mathrm{Ad} : G \times \mathfrak{g} \to \mathfrak{g}$ satisfies $\mathrm{Ad}(g)\tilde{X} = \frac{d\phi_g(h)}{dh}|_{h=e}\tilde{X}$, where $\tilde{X} \in \mathfrak{g}$.

Then, defining the extended control input $\bar{u} = \begin{bmatrix} u^\top & \dot{\alpha} \end{bmatrix}^\top \in \mathbb{R}^3$ and full rank (due to the transversality condition) matrix $\bar{C} = [C \ -A]$, formula (11) can be presented as follows

$$\nu = \mathrm{Ad}^X(f)\left(\bar{C}\bar{u} - A_t\right). \tag{12}$$

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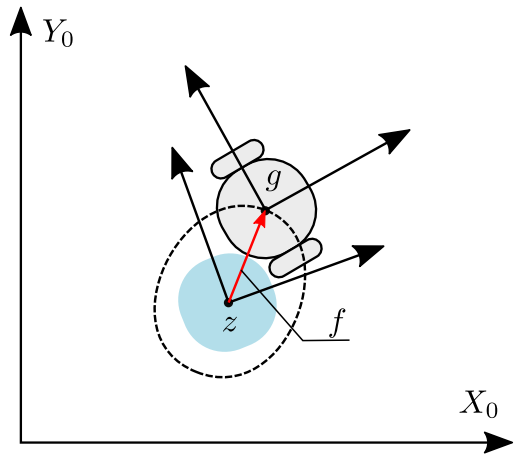


Fig. 1: Virtual frame z and the body frame g of the unicycle-like planar vehicle, illustrating their relative configuration in the planar workspace.

The essential feature of the considered approach is to couple motion of the nonholonomic unicycle with virtual omnidirectional kinematics. The assumed constraint (8) indicates that for the given z, g remains in the neighborhood of z, see Fig. 1. Therefore, the control problem can be recast as a control problem for a virtual system. Thus, controlling the motion of (4) it is expected that the system (5) approximates its motion with the desired accuracy. Taking into account (12), it can be concluded that for any bounded ν one can compute the following extended input

$$\bar{u} = \bar{C}^{-1}\left(\mathrm{Ad}^{X}(f^{-1})\nu + A_t\right), \tag{13}$$

which governs the trajectory $g(t)$ of the unicycle system (5).

C. Exciting transverse function

The choice of the transverse function f, which shapes the trajectory $g(t)$ in the neighborhood of $z(t)$, is essential for the main control task considered in this paper. Since the gradient of the measurement function μ around z is unknown and must be approximated, it is necessary to define a trajectory within this neighborhood that provides sufficient excitation for reliable gradient estimation.

We propose injecting an excitation component into the transverse function f. Motivated by the generalized construction of transverse functions on Lie groups [12], [14], we select a preliminary transverse function $\bar{f}$ for the unicycle and left-translate it by a time-dependent component h. To be more precise, the exciting transverse function can be defined as follows

$$f(\alpha, t) = h(t)\bar{f}(\alpha), \tag{14}$$

where

$$\bar{f}(\alpha) = \begin{bmatrix} \varepsilon_1 \sin(\alpha) \\ \varepsilon_1\varepsilon_2\varepsilon_3 \sin(2\alpha) \\ \varepsilon_2 \cos(\alpha) \end{bmatrix} \tag{15}$$

with ε_1, ε_2 and ε_3 are some coefficients selected to satisfy the transversality condition – for additional discussion see [13]. The term $h(t)$ is periodic, that is $h(t + T_e) = h(t)$, where $T_e > 0$. Specifically, we propose the following form of the excitation component

$$h(t) = \begin{bmatrix} \rho_1 \cos(\omega_e t) \\ \rho_2 \sin(\omega_e t) \\ 0 \end{bmatrix}, \tag{16}$$

where $\rho_1 \geq 0$ and $\rho_2 > 0$ are the design parameters, while $\omega_e = \frac{2\pi}{T_e}$ defines the frequency of oscillations.

Computing time derivative of (14) (the details are omitted here – one can follow computations referring to [14]) one obtains

$$\dot{f} = X(f)\left(A(\alpha)\dot{\alpha} + \mathrm{Ad}^{\mathrm{X}}(\bar{f}^{-1})w_h\right) \tag{17}$$

where

$$w_h = X^{-1}(h)\dot{h}. \tag{18}$$

Next, comparing (10) with (17) one concludes that $A_t = \mathrm{Ad}^{X}(\bar{f}^{-1})X^{-1}(h)\dot{h}$. As a result, the extended input (13) can be rewritten as follows

$$\bar{u} = \bar{C}^{-1}\mathrm{Ad}^{X}(f^{-1})\left(\nu + \mathrm{Ad}^{X}(h)X^{-1}(h)\dot{h}\right). \tag{19}$$

To gain insight into the behavior of the solution of (19), which governs the evolution of the exciting transverse function in the absence of the input ν, a series of numerical simulations was performed. The resulting motion patterns are illustrated in Fig. 2 for the following choice of parameters of $\bar{f}$: $\varepsilon_1 = 0.1$ m, $\varepsilon_2 = \frac{\pi}{3}$ rad and $\varepsilon_3 = 0.25$. In particular, the following excitation amplitudes (unit in meters) are considered:

E1: $\rho_1 = 0.4$, $\rho_2 = 0.4$ **E2**: $\rho_1 = 0.2$, $\rho_2 = 0.2$
E3: $\rho_1 = 0$, $\rho_2 = 0.4$ **E4**: $\rho_1 = 0$, $\rho_2 = 0.2$
E5: $\rho_1 = 0.2$, $\rho_2 = 0.4$ **E6**: $\rho_1 = 0.4$, $\rho_2 = 0.2$

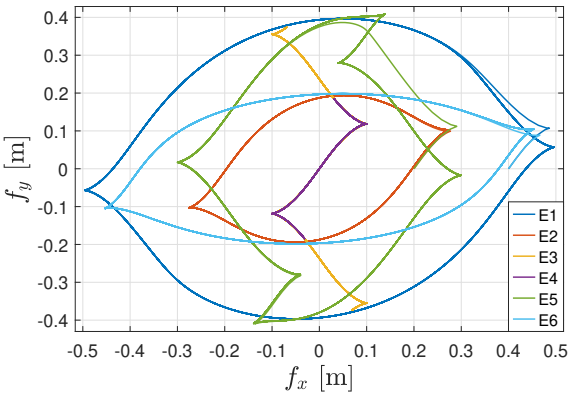


Fig. 2: Positional path of the transverse function f for selected parameter values defined at points E1–E6 obtained for $\nu \equiv 0$.

The obtained trajectories illustrate that the parameters of the transverse function significantly influence the motion patterns of the unicycle. These patterns enable continuous exploration of the objective function in the vicinity of z.

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D. Extremum seeking controller

Let us assume that $V(z) \in \mathbb{R}^3$ denotes a vector field governing the evolution of the virtual omnidirectional system (4), i.e., $\dot{z} = V$. Moreover, this vector field is determined by the gradient of the potential field μ such that

$$V = k(\nabla_z \mu)^\top = k \begin{bmatrix} \frac{\partial \mu}{\partial z_x} & \frac{\partial \mu}{\partial z_y} & 0 \end{bmatrix}^\top, \tag{20}$$

where $k > 0$ is a parameter scaling the vector field. In particular, only the position components are considered, while the angular motion is assumed to be irrelevant. Consequently, the trajectory z converges to a point $z^* \in G$ at which $\mu = \mu^*$. The last component of z is not taken into account since z_θ does not affect μ by assumption.

Next, consider V in the local frame of z, namely define

$$\nu = X(z)^{-1} V(z). \tag{21}$$

Regarding the extremum seeking problem, the gradient of μ is unknown and must be somehow restored. In this paper, we take advantage of the proposed form of the transverse function, which enables rendering of the neighborhood of the point z periodically.

To approximate the gradient at z we consider μ on the trajectory g within the time period T_e. Due to (8) this trajectory is equivalently represented by zf. Thus, $f(t)$ can be used to explore a neighborhood of z providing the special patterns – cf. paths presented in Fig. 2. As a result, μ can be measured locally along the trajectory provided by f. To approximate μ, it is assumed that, in the local coordinates defined by f, the function μ can be represented by a two-variable polynomial of at least second order, denoted as $\hat{\mu} : \mathbb{R}^2 \to \mathbb{R}$, parameterized by ψ. The problem to be solved can be formulated as an optimization problem and is defined as follows:

$$\hat{\psi} = \arg\min_p \int_\tau^{\tau+T_\mathrm{e}} \left(\hat{\mu}(f_{xy}(s), \psi) - \mu((zf(s))_{xy})\right)^2 ds, \tag{22}$$

under following assumption.

Assumption 4. *The excitation signal ensures persistent excitation over the estimation window T_e, providing sufficient richness of the data, and the approximation $\hat{\mu}(f_{xy}, \psi)$ is locally accurate in a neighborhood of z^*. Together, these conditions guarantee that the parameter estimate $\hat{\psi}$ reliably represents the local behavior of μ near the extremum.*

It is important to note that $\hat{\mu}$ is evaluated in local coordinates with respect to z, which in the considered case are represented by f. As a result, the following local approximation of the gradient can be considered

$$\widehat{\nabla}_z^{\mathrm{loc}} \mu = \left. \frac{\partial \hat{\mu}(f_{xy}, \hat{p})}{\partial f} \right|_{f=e}. \tag{23}$$

However, in the inertial frame, the gradient approximation becomes

$$\widehat{\nabla}_z \mu = (\widehat{\nabla}_z^{\mathrm{loc}} \mu) X(z)^\top. \tag{24}$$

Thus, recalling (20) and (21), we can design the control input for the virtual system as follows

$$\nu = k(\widehat{\nabla}_z^{\mathrm{loc}} \mu)^\top. \tag{25}$$

Since the quality of the gradient approximation is dependent on ν, the gain k has to be limited to maintain the stability of the closed-loop system $\dot{z} = k(\widehat{\nabla}_z \mu)^\top$ provided that the persistence excitation condition is met. To systematically enforce this condition as well as to limit magnitudes of robot velocities, the control input v can be redefined using the following scaling procedure

$$\bar{\nu} = \max\left\{1, \|\nu_{\max}^{-1} \nu\|_\infty\right\}^{-1} \nu, \tag{26}$$

under the following assumption

Assumption 5. *The saturation level $\nu_{\max} \in \mathbb{R}_+$ is selected sufficiently small to guarantee that the scaled control input $\bar{\nu}$ preserves stability of the closed-loop system.*

Substituting the nominal control input ν with $\hat{\nu}$ in (19) we can propose the following:

Proposition. *The control law* (19)*, together with the navigation rule* (25)*, the scaling procedure* (26)*, and the estimation algorithm solving* (22)*, ensures that the positional trajectory g_{xy} of the unicycle locally (i.e., within the domain of attraction) converges to a neighborhood of the maximizer p^*, provided that ρ_2 is chosen sufficiently large and k is chosen sufficiently small.*

By design, the proposed controller ensures that $z_{xy}(t)$ converges to p^*. However, due to the constraint (8), g_{xy} remains in the vicinity of p^*, with a radius determined by the parameters of the transfer function (14). Thus, the original control problem stated in Section III-A is solved.

IV. Simulation results

The intensity function was chosen in a quadratic form as $\mu(x, y) = -ax^2 - by^2$, with parameters $a = 1$ m and $b = 1$ m. The controller parameters were selected as $\varepsilon_1 = 0.1$ m, $\varepsilon_2 = \frac{\pi}{3}$ rad, and $\varepsilon_3 = 0.25$. Additionally, the control gain was set to $K = 0.03$ with a saturation limit $v_{\max} = 0.05$ m/s. The simulation study was conducted for three distinct initial conditions, two values of ω, and six different sets of excitation parameters.

Case I. The initial condition of the system was set to $g_0 = [3, -4, 0]^\top$ ([m, m, rad]). The excitation signal was defined by $\omega = 0.5$ $\frac{\mathrm{rad}}{\mathrm{s}}$ with amplitudes $\rho_1 = 0.3$ m and $\rho_2 = 0.3$ m.

Case II. The initial condition was modified to $g_0 = [3, -1, -\pi/2]^\top$, and the excitation frequency was increased to $\omega = 1$ $\frac{\mathrm{rad}}{\mathrm{s}}$, while keeping the amplitudes the same as in Case I, i.e., $\rho_1 = 0.3$ m and $\rho_2 = 0.3$ m.

Case III. The initial condition was set to $g_0 = [0, -4, \pi/2]^\top$, with excitation parameters $\omega = 0.5$ $\frac{\mathrm{rad}}{\mathrm{s}}$ and increased amplitudes $\rho_1 = 0.4$ m and $\rho_2 = 0.4$ m.

Case IV. The initial condition of the system was set to $g_0 = [3, -4, 0]^\top$ ([m, m, rad]). The excitation signal was defined by $\omega = 0.5$ $\frac{\mathrm{rad}}{\mathrm{s}}$ with amplitudes $\rho_1 = 0$ and $\rho_2 = 0.3$ m.

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Case V. The initial condition was modified to $g_0 = [3, -1, -\pi/2]^\top$, and the excitation frequency was increased to $\omega = 1 \frac{\text{rad}}{\text{s}}$, while keeping the amplitudes the same as in Case IV, i.e., $\rho_1 = 0$ and $\rho_2 = 0.3$ m.

Case VI. The initial condition was set to $g_0 = [0, -4, \pi/2]^\top$, with excitation parameters $\omega = 0.5 \frac{\text{rad}}{\text{s}}$, $\rho_1 = 0$ and increased amplitude $\rho_2 = 0.4$ m.

The simulation model was implemented in a hybrid manner. Specifically, the controller (19) and the unicycle kinematics (5) were integrated using the ode45 algorithm with MaxStep = 0.1 s, RelTol = 1e-3, AbsTol = 1e-6. While the external loop implementing (25), together with the solver computing the solution to (22), was updated at 10 Hz. Figures. 3 and 5 presents a three-dimensional graph that depicts the paths $g \in G$ overlaid on a surface plot illustrating the intensity distribution for three simulation cases. While Figs. 4 and 6 presents the time evolution of the configuration states, control inputs, and the intensity function during the extremum seeking control process for simulation Case I and Case IV respectively. All values shown in the plots are expressed in SI units. The results confirm that the controller correctly drives the system toward the maximum of the intensity function. It is also observed that the shape of the resulting trajectories may vary depending on the selected control parameters. Therefore, these parameters should be carefully tuned by the designer to achieve a fast system response while maintaining sufficiently accurate gradient estimation. One can observe, for example (cf. Figs. 4 and 6), that introducing additional excitation along the robot's x-axis through a nonzero parameter ρ_1 may improve the convergence rate by enhancing gradient estimation via the resulting closed-loop motion patterns. However, this ellipse-like trajectory introduces asymmetry in the angular velocity ω plot due to the additional rotations generated by this type of motion.

It is noteworthy that, once the robot reaches the vicinity of the maximum, the modulation mechanism introduced by (16) can be attenuated by making ρ_1 and $\rho_2 \to 0$ while simultaneously setting $k \to 0$ in (25). The latter prevents improper navigation when the result provided by (22) may be inaccurate because of insufficient excitation. An appropriate convergence criterion can be applied, for example, when the objective function value μ approaches its maximum or when the estimated gradient falls below a predefined threshold, thereby eliminating the robot's residual oscillations.

V. Conclusions

This paper presented an extremum seeking control approach for a nonholonomic mobile robot of unicycle type based on the transverse function framework. The proposed method is defined at the kinematic level (dynamics of the robot are not considered explicitly) incorporates an excitation component injected into the transverse function, which ensures persistent excitation and enables local gradient estimation. The simulation results demonstrate that the controller successfully drives the system toward the extremum of the objective function, confirming the validity of the approach. It has also been observed that the proposed excitation mechanism is effective in supporting the extremum seeking process.

Future work will focus on investigating different parameterizations of the transverse function and analyzing how they

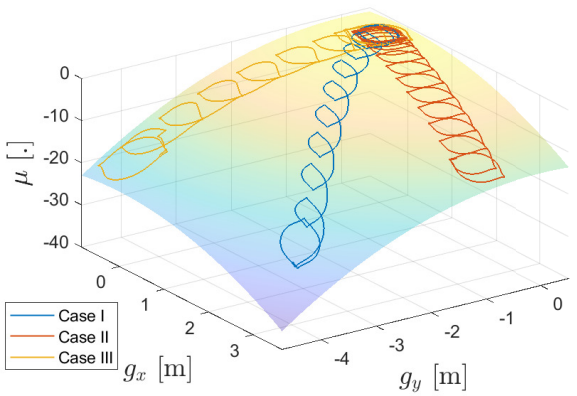


Fig. 3: Paths in the configuration space during the extremum seeking control for simulation Case I-III.

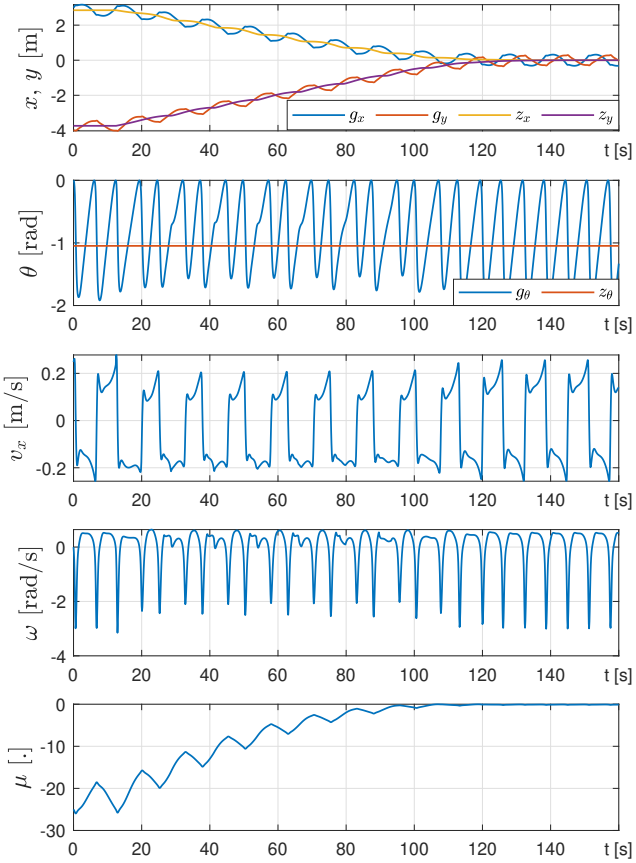


Fig. 4: Time plots of configuration g, z (position, orientation), control signal u (v_x, ω) and intensity μ during the extremum seeking control for simulation Case I.

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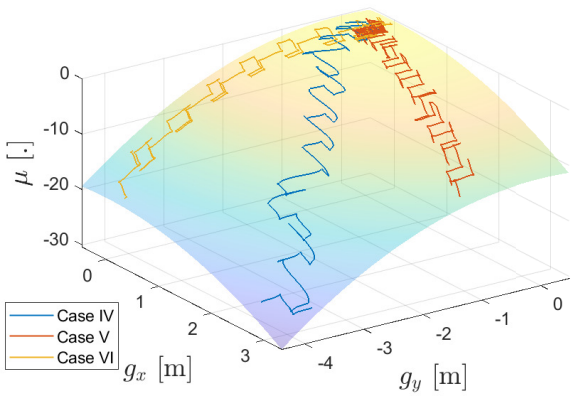


Fig. 5: Paths in the configuration space during the extremum seeking control for simulation Case IV-VI.

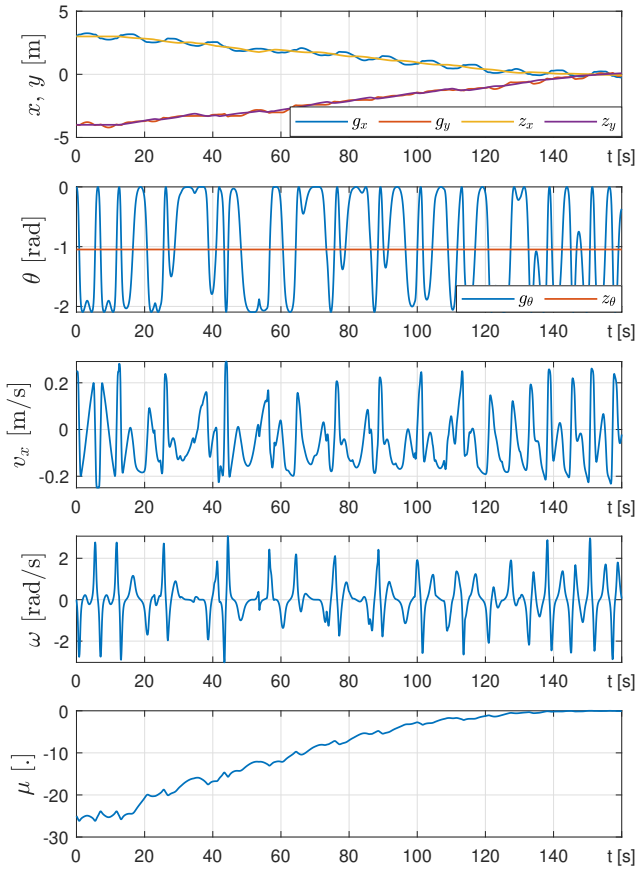


Fig. 6: Time plots of configuration g, z (position, orientation), control signal u (v_x, ω) and intensity μ during the extremum seeking control for simulation Case IV.

affect the performance of the proposed control scheme. It is also of interest to extend the method to more complex systems, such as car-like vehicles. Although the presented simulation results suggest that the proposed control approach may be applicable to real systems, further research is required to assess its robustness to measurement noise and time delays under realistic operating conditions.

In addition, a comprehensive comparison with alternative extremum seeking and optimization strategies is planned in order to evaluate the relative performance and practical applicability of the proposed approach. The controller structure also allows independent regulation of the unicycle orientation, which may enable an extension of the method to problems where the objective function is defined over a three-dimensional domain.

REFERENCES

[1] M. Krstic and K. Ariyur, *Real-Time Optimization by Extremum-Seeking Control*. Hoboken, New Jersey: John Wiley & Sons, 2003.

[2] C. Danielson, S. A. Bortoff, and A. Chakrabarty, "Extremum seeking control with an adaptive gain based on gradient estimation error," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 53, no. 1, pp. 152–164, 2023.

[3] G. Gelbert, J. P. Moeck, C. O. Paschereit, and R. King, "Advanced algorithms for gradient estimation in one- and two-parameter extremum seeking controllers," *Journal of Process Control*, vol. 22, no. 4, pp. 700–709, 2012.

[4] A. S. Matveev, H. Teimoori, and A. V. Savkin, "Navigation of a unicycle-like mobile robot for environmental extremum seeking," *Automatica*, vol. 47, no. 1, pp. 85–91, 2011.

[5] J. P. Laumond, S. Sekhavat, and F. Lamiraux, *Guidelines in nonholonomic motion planning for mobile robots*. Berlin, Heidelberg: Springer Berlin Heidelberg, 1998, pp. 1–53.

[6] S.-J. Liu and M. Krstic, *Stochastic Averaging and Stochastic Extremum Seeking*, 1st ed., ser. Communications and Control Engineering. Springer London, 2012.

[7] V. Grushkovskaya, A. Zuyev, and C. Ebenbauer, "On extremum seeking controllers based on the Lie bracket approximation in domains with obstacles," *Proceedings in Applied Mathematics and Mechanics*, vol. 18, no. 1, p. e201800298, 2018.

[8] V. Grushkovskaya and A. Zuyev, "Extremum seeking approach for nonholonomic systems with multiple time scale dynamics," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 5392–5398, 2020, 21st IFAC World Congress.

[9] A. S. Matveev, M. C. Hoy, and A. V. Savkin, "Extremum seeking navigation without derivative estimation of a mobile robot in a dynamic environmental field," *IEEE Transactions on Control Systems Technology*, vol. 24, no. 3, pp. 1084–1091, 2016.

[10] P. Morin and C. Samson, "A characterization of the Lie Algebra Rank Condition by transverse periodic functions," *SIAM Journal on Control and Optimization*, vol. 40, no. 4, pp. 1227–1249, 2002.

[11] ——, "Practical stabilization of driftless systems on Lie groups: the transverse function approach," *IEEE Transactions on Automatic Control*, vol. 48, no. 9, pp. 1496–1508, 2003.

[12] ——, "Control of nonholonomic mobile robots based on the transverse function approach," *IEEE Transactions on Robotics*, vol. 25, no. 5, pp. 1058–1073, 2009.

[13] D. Pazderski, "Application of transverse functions to control differentially driven wheeled robots using velocity fields," *Bulletin of the Polish Academy of Sciences Technical Sciences*, vol. 64, no. 4, pp. 831–851, 2016.

[14] ——, "Waypoint following for differentially driven wheeled robots with limited velocity perturbations," *Journal of Intelligent & Robotic Systems*, vol. 85, no. 3, pp. 553–575, 2017.

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